DATAOPS

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INNOVATION HACKATHON

AI-Powered Document Forensics and Contextual Intelligence System

1. Background

Financial institutions and enterprises process large volumes of documents, including application forms, identity proofs, transaction requests, declarations, and supporting documents.

Traditional document-processing systems primarily rely on Optical Character Recognition (OCR) to extract information and predefined validation rules to verify individual fields. However, extracting information does not necessarily establish whether the information is authentic, consistent, or reliable.

Documents may contain subtle alterations, overwritten fields, inconsistent information, manipulated images, misplaced signatures, or discrepancies that become apparent only when multiple documents are examined together.

Identifying such irregularities requires more than conventional OCR. It requires a system capable of understanding document structures, interpreting contextual information, examining visual characteristics, and establishing relationships between multiple sources of information.

This challenge aims to develop an **AI-powered document intelligence and forensic investigation system** that goes beyond conventional text extraction to identify, investigate, and explain document-level anomalies.

2. Problem Statement

Design and develop an AI-powered Intelligent Document Forensics System capable of analyzing, interpreting, and investigating digital and scanned documents to identify

suspicious modifications, contextual inconsistencies, and potential document manipulation.

The proposed system must go beyond conventional OCR-based text extraction. It should combine document intelligence, computer vision, contextual analysis, and intelligent verification mechanisms to establish the reliability of information contained within submitted documents.

Unlike conventional document verification systems that independently validate individual fields, the proposed solution should be capable of analyzing relationships between multiple documents, identifying discrepancies that may not be visible through isolated analysis, and investigating suspicious document characteristics.

The system must demonstrate the ability to:

- Extract and interpret information from scanned and digital documents.
- Understand document layouts, structures, and relationships between fields.
- Identify suspicious modifications and visual inconsistencies.
- Correlate information across multiple related documents.
- Identify contextual discrepancies and potentially manipulated information.
- Generate evidence-backed explanations for identified anomalies.
- Maintain traceability between identified findings and their original document sources.

The solution must demonstrate contextual understanding and evidence-driven investigation rather than relying exclusively on OCR output or predefined keyword-based rules.

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3. Expected User Experience and End-to-End Flow

The proposed system should support the following end-to-end document investigation workflow:

Document Submission → Document Classification → OCR & Information Extraction → Structural Understanding → Visual Forensic Analysis → Contextual Verification → Cross-Document Correlation → Anomaly Investigation → Evidence Generation → Investigation Dossier

Workflow Explanation

1. **Document Submission:** Accept scanned documents, images, and digital documents as input.
2. **Document Classification:** Identify the document type and determine its structure and relevant information fields.
3. **OCR and Information Extraction:** Extract printed and handwritten information wherever applicable.
4. **Structural Understanding:** Organize extracted information into structured representations while preserving its relationship with the original document.

5. **Visual Forensic Analysis:** Examine document characteristics and identify suspicious modifications, overwritten fields, inconsistent fonts, image alterations, and other visual anomalies.
6. **Contextual Verification:** Examine relationships between extracted fields and identify inconsistencies within individual documents.
7. **Cross-Document Correlation:** Compare information across multiple related documents to identify conflicting or inconsistent records.
8. **Anomaly Investigation:** Analyse identified discrepancies and establish their supporting evidence.
9. **Evidence Generation:** Highlight suspicious regions and associate findings with the corresponding original document locations.
10. **Investigation Dossier:** Generate a structured report containing identified anomalies, supporting evidence, contextual explanations, and areas requiring further verification.

The system should display processing progress, extracted information, identified anomalies, confidence levels, and supporting evidence.

It should also clearly distinguish between observed visual characteristics, detected inconsistencies, and potential manipulation.

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4. Technology Integration Scope

Participants may integrate technologies and frameworks from the following domains:

Technology / Domain	Primary Expected Role
OCR Engines	Extract printed and handwritten information from documents.
Computer Vision	Analyse document images and identify visual irregularities.
Image Processing	Examine image composition, alterations, and suspicious regions.
Document Layout Analysis	Identify document structures, sections, tables, and fields.
Natural Language Processing (NLP)	Interpret extracted information and identify contextual inconsistencies.
Large Language Models (LLMs)	Assist with contextual reasoning, investigation explanations, and report generation.
Machine Learning / Deep Learning	Identify patterns and anomalies in document information and visual characteristics.
Signature Analysis	Examine signature characteristics and their placement within documents.
Entity Matching	Establish relationships between information extracted from multiple documents.

Participants may use suitable open-source frameworks, pretrained models, APIs, and libraries.

Teams must clearly document the technologies used, their respective roles, and the limitations of their implementations.

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5. Functional Requirements

The proposed system must demonstrate the following functional capabilities.

5.1 Document Ingestion and Classification

- Accept scanned documents, images, and digital documents.
- Support multiple documents within a single investigation.
- Identify document types and relevant document structures.
- Maintain document identifiers and source references.

5.2 OCR and Information Extraction

- Extract printed and handwritten information wherever applicable.
- Identify relevant fields, sections, and tables.
- Convert extracted information into a structured representation.
- Maintain field-level confidence and references to original document locations.

5.3 Document Structure Understanding

- Identify relationships between different fields and sections.
- Recognize document layouts and information hierarchies.
- Identify missing, duplicated, or misplaced information.
- Preserve the relationship between extracted information and its original location.

5.4 Visual Forensic Analysis

- Identify suspicious alterations and overwritten fields.
- Examine inconsistencies in fonts, spacing, alignment, and document structure.
- Identify suspicious image insertions, replacements, or modifications.
- Analyse signatures and their placement within the document.
- Highlight potentially manipulated regions for further investigation.

5.5 Contextual Information Verification

- Identify inconsistencies between related fields within a document.
- Compare information across multiple submitted documents.
- Identify conflicting information and mismatched records.
- Detect missing, duplicated, or contextually inconsistent information.

5.6 Cross-Document Correlation

- Establish relationships between multiple documents belonging to the same application or transaction.
- Compare extracted information across related documents.
- Identify discrepancies that become visible only when multiple documents are analysed together.
- Maintain traceability between related documents and their extracted information.

5.7 Evidence Mapping and Investigation

- Associate each identified anomaly with its original document.
- Highlight suspicious regions within the document.
- Provide supporting visual and textual evidence.
- Explain why a particular field or region requires further investigation.
- Distinguish between confirmed observations and potential anomalies.

5.8 Investigation Dossier Generation

Generate a structured investigation dossier containing:

- Document overview.
- Extracted information.
- Identified inconsistencies.
- Visual forensic observations.
- Cross-document discrepancies.
- Supporting evidence.
- Confidence levels.
- Investigation explanations.
- Areas requiring additional verification.

5.9 Audit Trail

- Maintain records of document submissions and processing activities.
- Record identified anomalies and their associated evidence.
- Maintain investigation history.
- Preserve traceability between original documents, extracted information, and generated findings.

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6. Minimum Demonstrable Prototype vs. Advanced Extensions

Participants must distinguish between the mandatory capabilities required for a working prototype and additional advanced features.

Mandatory for Working Prototype	Advanced / Bonus Features
Document ingestion and OCR-based information extraction.	Advanced document classification and layout understanding.
Structured extraction of information from multiple documents.	Handwritten text recognition and multilingual document processing.
Identification of suspicious document regions and visual inconsistencies.	Advanced image forensics and manipulation localization.
Identification of inconsistencies within and across documents.	Intelligent cross-document relationship discovery.
Evidence mapping with references to original document locations.	Advanced contextual reasoning and anomaly correlation.
Generation of a structured investigation dossier.	Automated investigation summaries and interactive evidence exploration.
Interface for reviewing extracted information and findings.	Investigation history, analytics, and advanced audit workflows.

Prototype Expectations:

- The minimum demonstrable prototype must implement the mandatory capabilities listed above.
- Advanced features are optional and may be implemented to enhance the solution.
- Participants are not required to implement every technology or feature listed in the Technology Integration Scope.
- Teams should clearly document the capabilities implemented, partially implemented, planned, and unsupported.
- A recorded demonstration alone is not sufficient to establish that a particular capability is functional.

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7. Document Investigation Dossier – Required Sections

The generated investigation dossier must contain the following sections:

7.1 Document Overview

- Document identification.

- Document type.
- Source information.
- Processing status.

7.2 Extracted Information

- Extracted fields and values.
- Field-level confidence.
- Document structure.
- Original document references.

7.3 Contextual Analysis

- Identified field-level inconsistencies.
- Missing or duplicated information.
- Conflicting information across documents.
- Cross-document discrepancies.

7.4 Visual Forensic Findings

- Suspicious document regions.
- Potential alterations.
- Font, alignment, and structural inconsistencies.
- Image manipulation observations.
- Signature-related observations.

7.5 Evidence and Traceability

- Original document references.
- Highlighted regions.
- Supporting visual evidence.
- Associated extracted information.
- Confidence and observation status.

7.6 Investigation Summary

- Identified anomalies.
- Supporting evidence.
- Potential explanations.
- Areas requiring additional verification.
- Investigation limitations.

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8. Standards and Reference Frameworks

Participants may refer to the following standards and technical references wherever applicable:

1. ISO/IEC 27037 – Guidelines for identification, collection, acquisition, and preservation of digital evidence.
2. ISO/IEC 27041 – Guidance on assuring the suitability and adequacy of incident investigative methods.
3. ISO/IEC 27042 – Guidelines for the analysis and interpretation of digital evidence.
4. NIST SP 800-86 – Guide to Integrating Forensic Techniques into Incident Response.
5. OWASP Machine Learning Security Top 10, wherever applicable to AI-related security considerations.

Participants should use these references as applicable without claiming formal compliance solely through their implementation.

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9. Required Deliverables

Participating teams must provide the following:

1. **Working AI Agent / Prototype:** A functional document intelligence and forensic investigation system demonstrated using authorized sample documents.
2. **Architecture Overview:** An architecture diagram covering OCR, document processing, AI models, computer vision, contextual analysis, data flow, and evidence management.
3. **Sample Investigation Dossier:** An AI-generated investigation report containing extracted information, identified anomalies, and supporting evidence.
4. **Source Repository:** A source code repository containing setup instructions, dependencies, licensing assumptions, and execution instructions.
5. **Demonstration Dataset:** Genuine and artificially manipulated sample documents used to demonstrate the implemented capabilities.
6. **Technical Documentation:** Documentation describing the technologies, models, methodologies, and processing pipeline.
7. **Evidence Traceability Demonstration:** A demonstration showing how identified findings can be traced back to their original document locations.
8. **Limitations and Assumptions:** Documentation covering known limitations, assumptions, unsupported scenarios, and potential false positives.

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10. Judges' Live Demonstration Checklist

During the final demonstration, judges should evaluate the following:

- Upload multiple authorized sample documents and demonstrate document ingestion.
- Show document classification and OCR-based information extraction.
- Demonstrate how extracted information is converted into a structured representation.
- Show the identification and highlighting of suspicious document regions.
- Demonstrate the identification of inconsistencies within individual documents.
- Demonstrate cross-document comparison and identification of conflicting information.
- Show how the system establishes relationships between different findings.
- Inspect the evidence supporting identified anomalies.

- Demonstrate the generation of the final investigation dossier.
- Trace selected findings back to their original document locations.
- Demonstrate how the system distinguishes potential anomalies from confirmed observations.
- Demonstrate safe failure handling and identify unsupported capabilities honestly.

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11. Judging Criteria (100 Points)

The proposed solutions will be evaluated based on the following criteria:

Criterion	Points	What Judges Assess
OCR & Information Extraction	15	Accuracy, reliability, and structured extraction of document information.
Document Understanding	15	Interpretation of document layouts, fields, and relationships.
Visual Forensic Analysis	15	Identification of suspicious modifications and visual inconsistencies.
Contextual Intelligence	20	Identification of contextual discrepancies and cross-document inconsistencies.
Evidence & Traceability	10	Ability to associate findings with original document evidence.
AI Reasoning & Explainability	10	Quality of investigation explanations and anomaly reasoning.
Technical Implementation	10	Architecture, integration, functionality, and robustness.
Innovation	5	Novelty and practical applicability of the proposed solution.
TOTAL	**100**	

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12. Near-Perfect Solution Characteristics

A high-quality solution should demonstrate:

- Accurate and reliable information extraction.

- Strong document structure understanding.
- Effective visual anomaly identification.
- Context-aware document analysis.
- Strong cross-document correlation.
- Evidence-backed anomaly identification.
- Precise mapping of findings to original document regions.
- Clear distinction between observations and potential manipulation.
- Explainable investigation results.
- Reliable handling of multiple document formats.
- Transparent confidence and uncertainty.
- Professional investigation dossier generation.

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13. Safety, Ethics and Assessment Integrity

Participants must adhere to the following requirements:

- Use only authorized documents and datasets for testing.
- Do not use real customer or investor information without appropriate authorization.
- Protect sensitive information throughout document processing.
- Do not treat OCR output as verified ground truth.
- Do not automatically classify every detected anomaly as confirmed fraud.
- Clearly distinguish detected inconsistencies from confirmed manipulation.
- Ensure that automated findings are explainable and traceable.
- Preserve human oversight for consequential decisions.
- Maintain appropriate records of document processing and investigation activities.

Success Criterion

Judges should see a working intelligent document investigation system that goes beyond conventional OCR and field-level validation.

The solution should demonstrate its ability to understand document structures, examine visual characteristics, correlate information across multiple documents, identify contextual inconsistencies, and generate evidence-backed investigation findings.

The ultimate objective is to transform document processing from simple information extraction into intelligent, contextual, and evidence-driven document investigation.